MOTOR: Musculoskeletal Optimisation Through Open Robotics Simulation

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Working draft, 9 November 2026

Abstract

Physical AI now estimates and generates moving human bodies at a scale biomechanics has never had, yet the two fields cannot use each other's work: each is fragmented across its own body models, skeletons and solvers, and every bridge between them is a chain built for one body model and one route. We present MOTOR, a single open solver that takes any body physical AI produces, in SOMA's unified form of skin mesh, segment rotations and joint centres, drives the full-body musculoskeletal model MyoFullBody with it, and returns what biomechanics measures: joint angles in the ISB conventions, joint moments, ground reaction forces and the force in every muscle. It runs on physical AI's own GPU stack, Newton and MuJoCo Warp, because the more complex the bodies and movements we simulate, the more parallel a solver has to be. From one camera on OpenCap's LabValidation cohort, MOTOR's joint angle error is 5.16° (OpenCap Monocular: 4.8°) and its vertical ground force error from the 45° camera 11.6% of body weight (OpenCap, two cameras: 11.4%), in 66 s per trial on one GPU against OpenCap's 44 to 70 minutes; its joint moments have twice OpenCap's error. One solver in place of many chains gives physical AI bodies that obey the mechanics of muscle and bone, and gives biomechanics the scale to study the forces behind human movement far beyond the laboratory.

1 Related work

Physical AI, concerned with agents that perceive and act in the physical world, has produced a broad set of tools for the moving human body: video estimators [1, 2], motion generators [3] and GPU simulators such as Newton and MuJoCo Warp [4, 5], on body models that each have their own skeleton [6, 7]. SOMA [8] recovers one unified skeleton from the meshes of SMPL, SMPL-X, MHR and other body models, and GEM-X and Kimodo now output SOMA bodies directly, which makes SOMA the field's first standard body.

Biomechanics quantifies the forces that produce movement: from motion capture and force plates, musculoskeletal software such as OpenSim [9, 10] estimates joint moments and muscle forces [11], with models and solvers as varied as physical AI's body models were [12, 13]. MyoFullBody, from MyoSuite [14], is a candidate to serve musculoskeletal modelling as SOMA serves body models.

Existing bridges are each built for one body model and one route: OpenCap fits an OpenSim model to keypoints from smartphone video [15, 16], HSMR and BioPose regress biomechanical poses from images [17, 18], differentiable MuJoCo models are fitted to video [19], and MuscleMimic and MS-Human-700 train muscle-driven bodies on MuJoCo Warp [20, 21].

The Portable Biomechanics Laboratory (PBL) [22] fits a biomechanical model to monocular smartphone video end to end on MuJoCo's MJX, representing each trial's movement as an implicit function of time and optimising the body scale and marker offsets jointly across a session's trials, with joint angle errors below 3°. Biomechanical 3D Body (B3D) [23], from the same group, distils

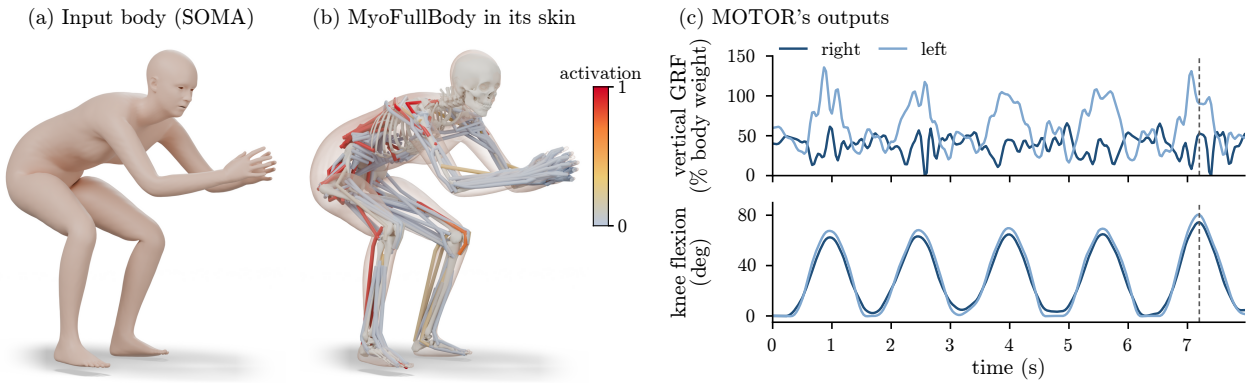


Figure 1: MOTOR on a squat, LabValidation's 45° camera. (a) GEM-X's SOMA body at the deepest point. (b) MOTOR's MyoFullBody in that skin, muscles coloured by activation. (c) MOTOR's vertical ground force per foot and knee flexion; dashed: (a) and (b).

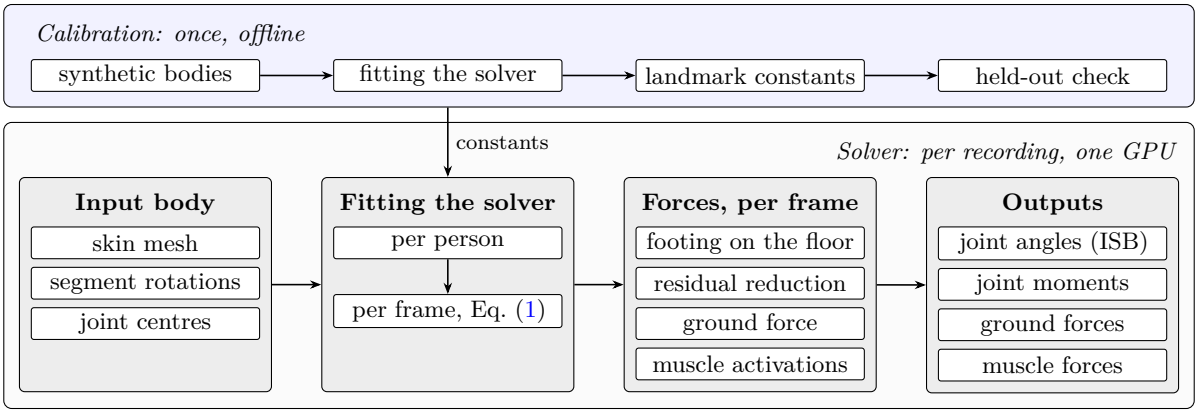


Figure 2: The structure of MOTOR: a calibration once, offline, and a solver per recording that fits and then solves the forces in every frame.

biomechanical pose into a per-frame regressor that comes within half a degree of PBL's lower-limb angle error on a clinical cohort (3.66 against 3.24°) at two to three orders of magnitude its speed. SKEL [24] places a biomechanical skeleton inside the skin of the SMPL body model. MOTOR takes from PBL and B3D its constants fitted once across many frames or bodies and its GPU computation, and, like B3D, solves each frame on its own. It adds a SOMA body as input, SKEL's bones inside the skin as a fitting target, and ground reaction and muscle forces solved on the same GPU.

MOTOR is a single solver between the two standards, from a SOMA body, and through SOMA any body model it unifies, to MyoFullBody's angles in the ISB conventions [25, 26] and its forces. To our knowledge, it is the first open chain from video to muscle forces (fit, residual reduction, static optimisation) on Newton and MuJoCo Warp.

2 Method

MOTOR has two layers (Figure 2; Appendix A in full): a calibration, run once and offline, and a solver per recording. MyoFullBody [14, 27] has 416 muscles and 132 joints; its bones are sized to the SOMA body's segments and its masses distributed by their volumes.

Calibration. The constants that hold for every body (each landmark's offset from its bone and weight, each segment end's position, two global weights) are estimated by one bounded least-squares

problem over synthetic SOMA bodies of several shapes and movements, as PBL estimates marker offsets across trials [22]. It minimises the bones' depth outside the skin and each segment's angle to its bone group, never a reference angle, and is accepted only when held-out bodies score within 10% of those used.

Fitting the solver. The solver first fixes the person's constants (each landmark's point on its bone and each joint inside a bone group) from about 30 of their frames together, as PBL and B3D do [22, 23], then fits MyoFullBody's coordinates q in each frame by

$$\begin{aligned} &\min_{q} \frac{1}{N_g}\sum_{g=1}^{N_g}\theta_g(q)^2 \;+\; \lambda_m \sum_{m} \bar{w}_m \left\| p_m(q) - \hat{x}_m \right\|^2 \;+\; \frac{\lambda_s}{N_b}\sum_{b=1}^{N_b}\left\| d_b(q) \right\|^2 \\ &\text{s.t. } q_{\mathrm{F}} = f(q_{\mathrm{L}}), \qquad q_{\min} \le q_{\mathrm{L}} \le q_{\max}. \end{aligned} \tag{1}$$

Here θ_g is the angle between bone group g's orientation and its SOMA segment's rotation, composed with a fixed rest alignment read from the two models at rest, each goal normalised by its count as in OpenSim's inverse kinematics [10]; each marker m, a landmark or segment end, pulls its model point p_m toward its point $\hat{x}_m$ on the SOMA body; and d_b is bone vertex b's offset outside the SOMA skin, after SKEL [24]. Coupled coordinates q_{F} follow the leading q_{L} through MyoFullBody's polynomials f, within published ranges.

Forces. MOTOR stands the fitted path on the floor the video shows, a plane fitted to the depth estimator MoGe-2's points [28], moving the body only up or down; the ground pushes only on points of the SOMA skin that touch it. A residual reduction in the manner of OpenSim's RRA [9] adjusts the pelvis, trunk and masses within a per-frame cap so that the ground can explain the movement. The forces are then split in the two stages of OpenSim's static optimisation, which keeps the problem well posed: the ground reaction force from the pelvis' six balance equations alone, then the 416 muscle activations with that force fixed. What the pelvis still needs is the residual, judged against 2% of body weight and 1% of weight times centre of mass height [29] and never applied. Every step solves all frames at once on the GPU, with Warp [30] and MuJoCo Warp [5] inside Newton, and two runs agree bit for bit [31].

3 Results

In our pipeline GEM-X [1, 32] turns one phone video into a SOMA body and MOTOR solves it. OpenCap's LabValidation cohort [15] has nine healthy subjects filmed by five phones walking, squatting, rising from a chair and jumping, with motion capture, force plates and EMG; each of its 720 single-camera recordings is run on its own, 715 to the end, and subjects 7 to 11 never chose a setting (Appendix B). We compare with OpenCap, the only comparable state of the art, through its published numbers from two cameras [15] and, for angles, OpenCap Monocular [16]: same cohort and reference, not rerun by us. We did not rerun PBL on LabValidation: its code is AGPL-3.0, but the pretrained MeTRAbs keypoint models [33] it depends on are licensed for non-commercial use only.

On the matched set, where OpenCap Monocular reports, MOTOR's joint angle error is 5.16° (95% CI 4.77 to 5.59°; Table 1), 5.12° on the held-out subjects and 5.35° over all inputs. The vertical force under each foot follows the plates (Figure 3), with median correlations of 0.78 (right) and 0.79 (left) on the held-out subjects. Summed over both feet, its error from the 45° camera is 11.6% of body weight beside OpenCap's 11.4% from two cameras, and over all inputs smallest

Table 1: MOTOR beside OpenCap's published numbers. Angles on the matched set (45° camera, no drop jumps), the rest over all inputs; force in % body weight, both feet; moments in % weight × height.

	OpenCap, 2 cameras	OpenCap Monocular	MOTOR, 1 camera
Joint angle MAE (centred), matched 15 (°)	4.5	4.8	5.16 (3.06)
Ground force MAE, vert. / ant. / med.-lat.	11.4 / 3.5 / 3.8	–	16.2 / 7.4 / 3.0
vertical, 45° camera, no drop jumps	–	–	11.6
Joint moment MAE, OpenCap's 15	1.2	–	2.41
Vasti symmetry AUC vs EMG label (squats)	0.83	–	0.66
Time per trial	44–70 min	–	66 s

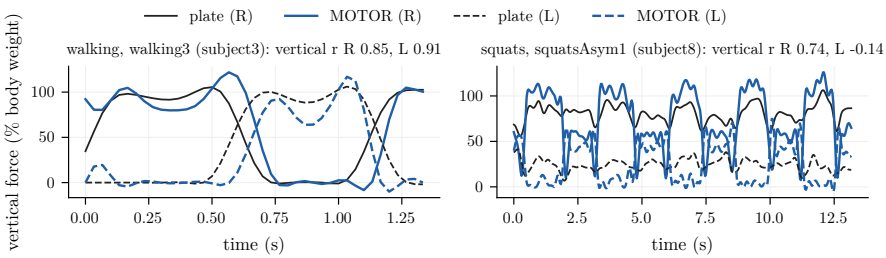


Figure 3: MOTOR's vertical ground force per foot against the plates (45° camera).

in squats (6.3%) and largest in walking and drop jumps (19.6 and 19.3%); the anterior force has twice OpenCap's error. The joint moments, 2.41% of weight times height against OpenCap's 1.2%, are 1.55 to 1.88% without drop jumps; the activations follow the EMG best in the vasti (r 0.36 to 0.42). OpenCap's recorded solves take 2635 to 4196 s, MOTOR a median 66 s on one L4 GPU (Appendix C).

4 Discussion

MOTOR's angles come within 0.4° of OpenCap Monocular's, whose input-quality gate, B3D found [23], rejected 400 of 1693 clinical inputs, among them posterior and self-occluded views; MOTOR scores every recording that ran. B3D's per-frame regressor comes within half a degree of PBL's whole-trial optimisation, and both results suggest that optimising the whole trial buys little kinematic accuracy for its cost. Our own whole-trial solve, which we leave to future work, bears this out. Built on PBL's structure with OpenCap's smooth contact [15] and the muscles solved with the movement, it held the pelvis residual within both limits on twice as many matched frames (40 against 20%), but no accuracy measure improved beyond its confidence interval (angles 5.29°, vertical force 12.5%, moments 2.67%, EMG r 0.15), at five times the time (309 against 66 s).

The joint moments have twice OpenCap's error, and the pelvis residual stays above its limits (medians 7.0% of body weight and 8.9% of weight times centre of mass height). The elbows come out 21 to 24° straighter than the laboratory's, following GEM-X's own elbow, and the pelvis tilt is offset by 5.3°. The activations agree only weakly with the EMG (mean r 0.23; rectus femoris −0.08) and can differ between the sides of a symmetric movement: in the squat of Figure 1 one calf is active and not the other, following the ground force's split between the feet (90 and 53% of body weight at the frame shown). Only GEM-X's and Kimodo's SOMA bodies have been run, and LabValidation has only healthy adults.

Markerless methods such as OpenCap removed the laboratory from estimating the forces behind movement; MOTOR replaces the bespoke chain with one solver between physical AI's standard body and a candidate standard musculoskeletal model.

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A The method in full

This appendix sets out the method in the order in which it runs (Figure 2, Section 2). The input body and the model are described first, then the correspondence between them and the fit that both later stages share. The calibration, done once and offline, follows, and finally the solver that runs on every recording: the person's constants, the per-frame fit, and the forces solved on the fitted path. Figure 4 shows the fitted model inside the skin, at full size, in two poses.

A.1 The input body

Any source that provides a skin mesh, segment rotations and joint centres can supply MOTOR's input, although only SOMA bodies, from GEM-X and Kimodo, have so far been run. In the validation the source is GEM-X [1, 32] applied to one smartphone video. GEM-X gives a SOMA body [8] per frame, consisting of its identity (shape), root translation and global segment rotations, at 30 Hz, the rate to which GEM-X resamples every clip. MOTOR takes this body as given and never changes its shape or pose. GEM-X is given the camera's field of view, which MoGe-2 [28] estimates from the same video, and its own contact post-process holds the standing feet still. Because GEM-X's output has no floor, the floor is estimated from the video: a plane is fitted to MoGe-2's points of the floor beside the person, the whole clip is rotated rigidly so that this plane is level, and the clip is lowered until the lowest point of its feet touches it.

A.2 The musculoskeletal model

The model is MyoFullBody from MyoSuite [14], its arm and hand taken from MoBL-ARMS [27, 34]. It has 416 muscles, 86 bones and 132 joints, of which one is the free root, 36 are coupled to another joint by a polynomial (those of the knee and the shoulder girdle, for example) and 95 move freely. Its bone groups (below) are sized to the person's SOMA body, each group's length taken from its SOMA segment, joint to joint, so that MOTOR solves for no length. Its masses are SOMA's segment volumes distributed over the subject's measured mass.

A.3 Segments, bone groups and landmarks

Segment rotations and bone groups. Each SOMA segment is coupled to a *bone group*, the bones of MyoFullBody that cover the same part of the body, one of which, the group's *frame bone*, carries the segment's rotation (Appendix D). A spine segment is coupled to the vertebrae at its height, paired by their share of the spine's height, and the hand to the bones of the wrist and hand, with the lunate carrying its rotation. Each segment drives one group. SOMA has two neck segments where MyoFullBody has a single rigid neck; the first drives it, and the second drives nothing, although its rotation error is still reported.

How a segment's frame lines up with its group's frame is read once from the two models at rest (SOMA's template in its bind pose, MyoFullBody in its default pose with its couplings applied) and kept as a fixed *rest alignment* per group. Each group's anatomical axes are constructed on both models from the same anatomy: the line between the group's end joints or between landmarks, the bone's own axis, or a hinge line, such as the line through the first and fifth metatarsal heads that the foot and toes share. The feet are then turned about that hinge line until SOMA's foot, flat on its floor at rest, corresponds to MyoFullBody's heel bone flat in its default pose, and the groups whose bones would stand outside SOMA's skin along those lines (the head first) are turned and shifted rigidly at rest to their least depth outside it. The rest alignment is therefore geometry of the

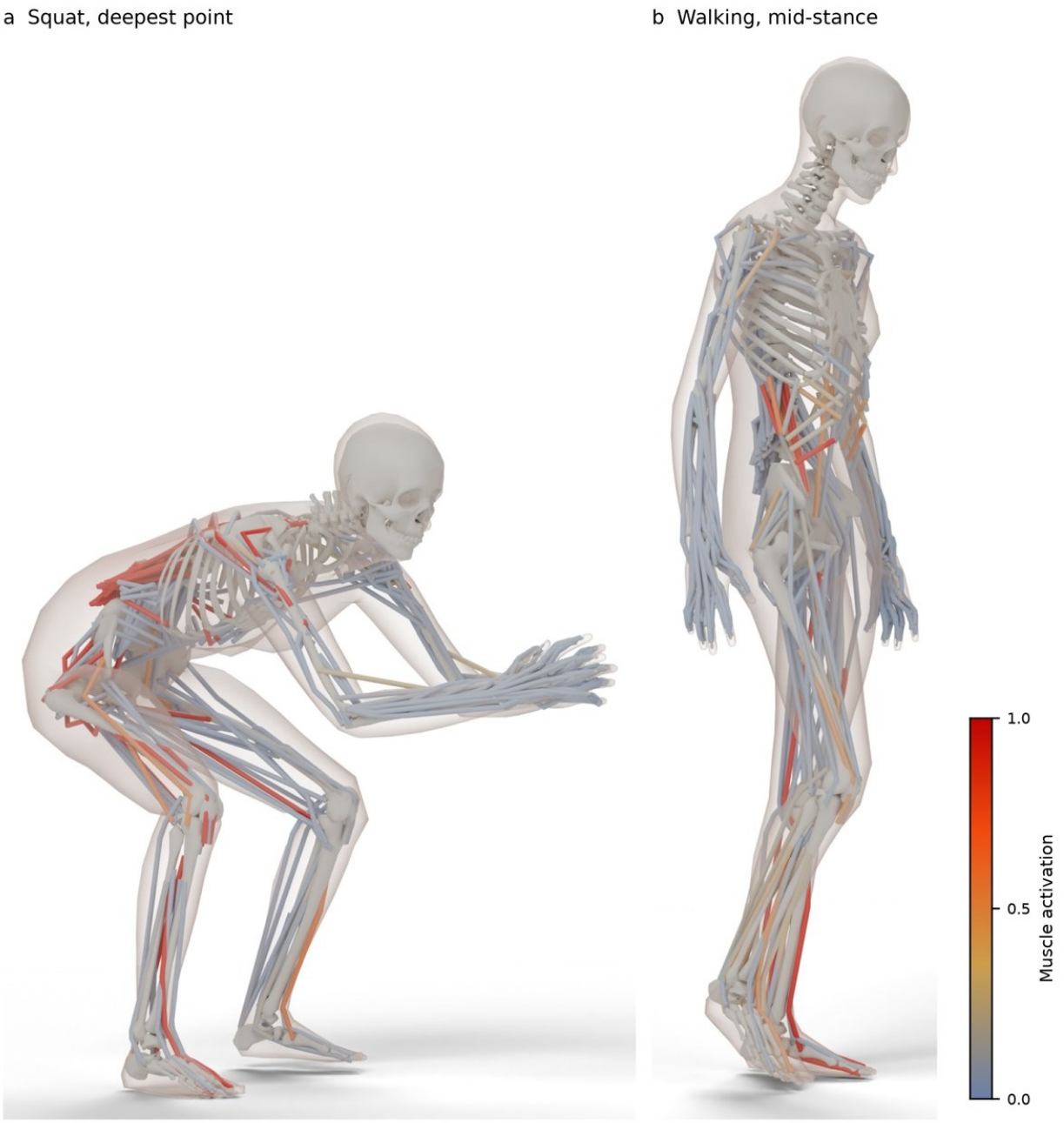


Figure 4: MyoFullBody fitted inside a person's skin from one LabValidation video: (a) a squat at its deepest point, (b) walking at mid-stance. Drawn from MOTOR's saved LabValidation results, the muscles coloured by MOTOR's activations.

two models: it is never fitted to motion data or to any reference joint angle, and never per person or per clip.

Virtual markers on anatomical landmarks. Beside the rotations, the fit uses virtual markers on anatomical landmarks, each defined by its anatomy following the ISB recommendations [25, 26]: the acromion, the humeral epicondyles, the anterior and posterior superior iliac spines, the malleoli, the metatarsal heads and others. Each landmark sits on a fixed point of the SOMA mesh, given as a face and barycentric weights, so that it moves with the person's skin, and on the matching point of a MyoFullBody bone, plus an offset from the bone toward the skin. A study of the two models placed 102 landmarks, of which the fit uses 84, each with its own weight. No licensed dataset's marker list is used.

Joint centres as segment ends. SOMA's rig joints mark where each bone group begins and ends: the hips, knees, ankles, shoulders, elbows, wrists and the joints between spine segments. Each such end acts as one more marker, placed at MyoFullBody's own joint centre there and aimed at SOMA's joint. The joints inside a group (the lumbar levels, the shoulder blade, the wrist bones and the joints of the inner foot) are aimed at nothing; the landmarks and the model's couplings place them. No joint is treated as a special case.

The skin. The bones belong inside the body from which they came. SKEL builds this idea into a body model, a biomechanical skeleton placed inside the skin of SMPL [24]; MOTOR instead uses the input's own skin as a target. Each frame's SOMA skin is posed, and a fixed subset of the vertices of every bone mesh is measured against it: each vertex's offset to the closest point of the skin, or zero when the vertex lies inside, as decided by the generalised winding number [35].

A.4 The fit

Fitting the solver, Equation (1), is the objective that the calibration and the solver on every recording share. For one frame, with the person's constants fixed, it finds MyoFullBody's coordinates q that come closest to the input body while every joint stays within its anatomical limits. The first term is the main one: for each of the N_g bone groups, it measures the angle between the frame bone's orientation $R_g(q)$ and its SOMA segment's rotation $\hat{R}_g$ composed with the group's rest alignment. The second pulls each marker m (the landmarks and the segment ends) at its point $p_m(q)$ on the model toward its point $\hat{x}_m$ on the SOMA body, weighted by $\bar{w}_m$, its weight w_m over the sum of all. The third holds the N_b bone vertices inside the skin, d_b being a vertex's offset outside it. Each goal is normalised by its own count, as OpenSim's inverse kinematics weighs its goals [10]. The markers' goal λ_m is set by the calibration (2.1169), and the skin goal λ_s is 10^3, chosen in a sweep that traded the bones outside the skin against the rotation error. A coupled joint (a *follower*) moves with a *leading* coordinate through MyoFullBody's own polynomial f, held as an equality constraint, and the limits are the published ranges of the leading coordinates, held as bounds. The fingers are held at SOMA's rest.

The reference solver is mink [36], a differential inverse-kinematics library on MuJoCo. Each of its steps solves one quadratic programme, in which the segment rotations and the marker positions are tasks on frames of the model, the couplings are equality constraints, the leading coordinates' ranges are limits and the finger joints are frozen; the step length is controlled as in Levenberg–Marquardt. The skin term adds three rows per bone vertex, its offset from the closest point of the skin, whose Jacobian keeps only the motion away from the closest face, edge or vertex of the skin mesh. On the GPU the same step is a damped Gauss–Newton step in the leading coordinates, solved for every

frame at once with Warp's tile Cholesky factorisation [30]. On every frame of a development clip it matches mink within 0.0025° in every joint angle and 0.0042 mm in every model point. A frame that does not converge in 300 steps stops the run with an error rather than being kept silently.

A.5 Calibration, once and offline

The calibration (level 1 in the code) determines the constants of the landmarks that hold for every body, and it runs once, offline, before the solver is applied to any recording. One bounded least-squares problem tunes, all together, each landmark's offset from its bone (for a landmark that defines a joint, only along its bone); each landmark's weight, mirrored and shared between left and right; each segment end's position on MyoFullBody, within 40 mm of the model's own joint centre; and two global weights, the markers' goal λ_m and the segment ends' weight among the markers. Its objective has two terms, each normalised by its value at the start so that the two weigh alike: each body region's bones' depth outside the SOMA skin, and each segment's rotation error against its bone group. Each evaluation runs the person stage and the per-frame fit on every tuning body.

The bodies are synthetic SOMA bodies (tall, short, heavy, slim, long- and short-legged), each performing seven clips (a Kimodo walk, a rig walk, arm raises, a reach, a squat, running and sit-to-stand). They are split by body into tuning and check bodies, with every clip of a body on one side. The check bodies are scored during the run but never used to choose, and a tuned point is accepted only when they score within 10% of the tuning bodies. The calibration's gradient is taken through the person stage and the per-frame fit by implicit differentiation; the columns that path does not yet cover are taken by finite differences, and every column is checked against finite differences before a run is used. The calibration thus sees only synthetic bodies' skin depth and rotation error, never any reference joint angle.

A.6 The solver, per recording

The person's constants (level 2). With the calibration fixed, the person's constants are solved from all of their frames together [23]. About 30 frames, spread over the person's clips in proportion to their length, are posed, and one sparse least-squares problem solves, together with those frames' coordinates, each landmark's point on its bone and the position of each joint inside a bone group, each within 3 cm of where the calibration and the model placed it and drawn back toward it by a weight. Only what the data can distinguish is solved: a joint position whose effect on the residuals is a combination of the other unknowns' effects stays where the model has it. The landmarks on the soles' pads and on the skull never move, so that a foot cannot sink through the sole of the skin nor the skull through the head. This stage has no skin term, because with one the solve moved the inside joints by 40 to 52 mm to reduce the bones' depth outside the skin instead of fitting the person. The constants are written into the person's model file, so that every later solve uses them.

The per-frame fit (level 3). With the person's constants fixed, every frame solves only its pose, by Equation (1): the rotations as the main term, the markers beside them, the bones inside the skin, and the joint limits and couplings as hard constraints. The path is then low-passed by a zero-lag fourth-order Butterworth filter at 6 Hz [11] before it is differentiated.

A.7 Forces on the fitted path

Footing on the seen floor. The clip stands on the floor seen in the video, levelled and lowered onto it as described for the input body. Where GEM-X's world drifts vertically between flights, the still feet's lowest sole is held at one height. Both steps move only the root up or down; the joint

angles are unchanged. A sit-to-stand's 40 cm chair [15] and a drop jump's box are drawn as ground where the clip's seated skin and first standing feet place them.

Contact. The ground pushes only on points of the person's SOMA skin, carried by the bones each segment rides, that lie within 5 mm of it or under it; MyoFullBody's own foot colliders are not used, because they stood inside the skin. A foot the video shows standing, by GEM-X's static contact on its ankle or toe, is pushed from its own lowest skin point when that point is at most 5 cm above the floor, because GEM-X's standing feet can stand a few millimetres to centimetres off the floor where its leg lengths and foot shape miss. This height, the 5 mm gap and the weight with which the ground force is shared over the touching points are the only contact settings, chosen on subjects 2 to 5 against their force plates, never against joint angles (Appendix E).

Residual reduction. Inverse dynamics on the MuJoCo model that Newton builds gives the generalised forces that each frame's motion needs. A residual reduction in the manner of OpenSim's RRA [9] then adjusts the fitted path's pelvis and trunk (the root's six coordinates and the lumbar leading coordinates) and the body's masses and torso centre of mass, with RRATool's own formulas, so that the ground explains the movement. It runs three rounds, and each round's move is taken only when the measured residual, lumbar and move cost falls. The move is capped on every frame: every segment's rotation within 5° and the pelvis within 2 cm of the fitted path, all rounds together, and the mass within 2% of the mass it was made with. The move is low-passed by construction, the feet are kept where they stood by re-solving the legs, and a frame whose move puts its bones further outside the skin has its share of the move lowered.

Static optimisation in two stages. The forces are split as OpenSim's static optimisation splits them, in two stages. The first gives the ground reaction force from the pelvis' six balance equations alone, shared over the touching skin points by a small weight on each newton, in two rounds. The second, with that force fixed, gives the joints' range stops and the activations of all 416 muscles at least squared activation, through MuJoCo's own muscle lengths, moment arms and force curves. Each knee's small rotation of the shin on the femur, which MyoFullBody's knee extensors reach only through moment arms of a few millimetres, is held by its ligaments as a stop, so that it does not decide how the vasti share their load. Solved as one least-squares problem, the split was ill-posed: a 10^{-6} nudge of the contacts moved an activation by up to 1.0, against at most 5×10^{-3} in two stages. What the pelvis still needs is the residual and what a joint still needs its reserve; both are recorded and judged, and never applied to the body.

One GPU, repeatable. Every step solves every frame at once on the GPU with MuJoCo Warp inside Newton, in Warp's deterministic mode, and two runs agree bit for bit.

B Validation protocol in full

Camera input. The video comes from five synchronised iPhones, Cam0 to Cam4 in LabValidation's numbering. The 144 non-static trials, each seen by five cameras, give 720 single-camera inputs, every one run on its own. For the matched comparison we use the 45° camera (Cam1) on the non-jumping activities and the fifteen joint angles that have a counterpart in OpenCap Monocular's set (Appendix B), the envelope on which OpenCap Monocular reports its results [16] and on which the author's earlier work reported GEM-X's angles [37]. The full technical validation pools all 720 inputs, every activity and every camera.

Metrics. Because video and reference are not synchronised at the millisecond level, each trial receives one time shift of the reference before scoring. Joint angles are scored by mean absolute error (MAE), centred MAE (cMAE, with each trial's constant offset removed, which separates tracking from offset), root mean square error and Pearson's r, with a 95% confidence interval from a per-subject bootstrap of 20,000 draws. Ground reaction forces are scored as OpenCap scores them, in % of body weight on three axes with both feet summed, and per foot by the correlation of the vertical force. Joint moments are scored in % of weight times height over OpenCap's fifteen, with the knee adduction moment's early-stance peak reported separately. In squats, the vasti symmetry index is scored by its AUC against the EMG's own asymmetry label. Activations are scored by on–off agreement and correlation with each EMG channel, and the pelvis residual by the share of frames within both limits.

Time. The solve time is MOTOR's wall time per trial on one named GPU, from the input body to its output, set beside OpenCap's recorded solves in its LabValidation release, whose hardware is not recorded.

Joint definitions. The matched comparison scores fifteen joint angles, the coordinates that have a counterpart in OpenCap Monocular's set: ankle angle, knee angle, hip flexion, hip adduction and hip rotation on both sides, together with pelvis tilt, pelvis list, lumbar extension, lumbar bending and lumbar rotation. The full technical validation also scores the shoulders (arm flexion, adduction and rotation), the elbows, the forearms' pronation–supination, the subtalar and metatarsophalangeal angles and pelvis rotation (Appendix C). Each angle is read from MyoFullBody's own coordinates after the fit, and nothing is calibrated or regressed against the reference. The hip, knee, ankle, subtalar, toe and elbow angles are MyoFullBody's own coordinates of the same name. The pelvis and lumbar angles are OpenSim's intrinsic Z, X, Y angles of the model's own body frames, as the lab's model defines them, and the shoulders are the same angles of each humerus on the torso. Pronation–supination has its zero at a different forearm pose in each model, so it is scored only by cMAE and r. The scoring follows the author's earlier work [37]: one sub-frame time shift of the reference per trial, found on the centred residual of both sides' hip, knee and ankle flexion; MAE, cMAE, RMSE and r; a dagger where the reference moves less than 15°; and the per-subject bootstrap of 20,000 draws.

C Full technical validation of MOTOR

This appendix follows the layout of Appendix B of [37]. It covers MOTOR's run of 7 November 2026 on all 720 single-camera inputs, with every activity including the drop jumps and each camera as its own input: 715 were scored, and Section C.9 gives the rest by cause. It reports every joint angle MOTOR outputs, the ground reaction forces on all three axes per activity, the joint moments, the residuals and every EMG channel, per activity and per camera. Every number is from paper/motor/evidence/labval-final-2026-11-07.txt, except the published solve times of Table 10 and where a comment names another source.

C.1 Every joint angle

Table 2: MAE and centred MAE (degrees) and mean r per coordinate (†: the reference moves less than 15°). Envelope: 45° camera, no drop jumps (89 trials). All: every scored input (715). The matched 15 first; pro_sup has no shared zero, so only its cMAE.

Coordinate	Envelope MAE	Envelope cMAE	Envelope r	All MAE	All cMAE	All r
ankle angle r	6.39	3.19	0.89	5.96	3.92	0.92
ankle angle l	4.61	3.32	0.89	5.51	3.75	0.92
knee angle r	6.06	3.04	0.99	5.99	3.17	0.99
knee angle l	5.94	3.24	0.98	6.09	3.50	0.98
hip flexion r	8.03	3.82	0.98	8.89	4.11	0.98
hip flexion l	7.37	3.53	0.98	8.93	4.17	0.98
hip adduction r	4.20	2.73	0.52†	3.72	2.59	0.52†
hip adduction l	4.72	2.76	0.60†	3.52	2.52	0.57†
hip rotation r	4.07	2.62	0.43†	4.36	2.77	0.51
hip rotation l	4.41	3.06	0.43†	4.20	3.00	0.54
pelvis tilt	6.06	3.64	0.51	7.02	4.39	0.40
pelvis list	3.15	2.46	0.17†	3.11	2.24	0.16†
lumbar extension	4.56	3.11	0.64	5.32	3.35	0.75
lumbar bending	4.19	3.21	0.44	3.69	2.54	0.45†
lumbar rotation	3.67	2.12	0.58†	3.97	2.71	0.44
arm add l	3.98	2.34	0.70	6.58	5.12	0.70
arm add r	4.23	2.23	0.51†	6.16	4.40	0.64
arm flex l	8.03	3.19	0.93	12.46	8.59	0.90
arm flex r	7.00	3.19	0.93	11.20	8.52	0.90
arm rot l	8.01	4.50	0.64	12.35	9.78	0.67
arm rot r	7.03	4.09	0.67	11.18	8.61	0.68
elbow flex l	23.68	4.61	0.74	24.70	5.67	0.79
elbow flex r	21.12	3.00	0.80	22.74	5.76	0.76
mtp angle l	15.35	2.47	0.15†	15.74	2.19	0.10†
mtp angle r	15.22	2.55	-0.02†	14.99	2.06	0.00†
pelvis rotation	2.70	2.57	0.50†	2.63	2.42	0.48†
pro sup l	-	2.82	0.16†	-	3.61	0.25
pro sup r	-	2.55	0.18†	-	3.46	0.26†
subtalar angle l	9.24	6.88	-0.21	9.73	6.25	-0.07
subtalar angle r	9.16	6.26	0.04	11.39	6.21	0.04
matched 15, pooled	5.16	3.06		5.35	3.25	

The elbows are about 21 to 25° off with a centred error of 3 to 6° (straighter than the lab's in each of the 89 envelope trials; GEM-X's arm estimate, which the fit follows), and the toes (mtp) 15° off with a centred 2 to 3°: constant offsets, not tracking errors. Figure 5 shows the matched 15, and Figure 6 the hip, knee and ankle waveforms.

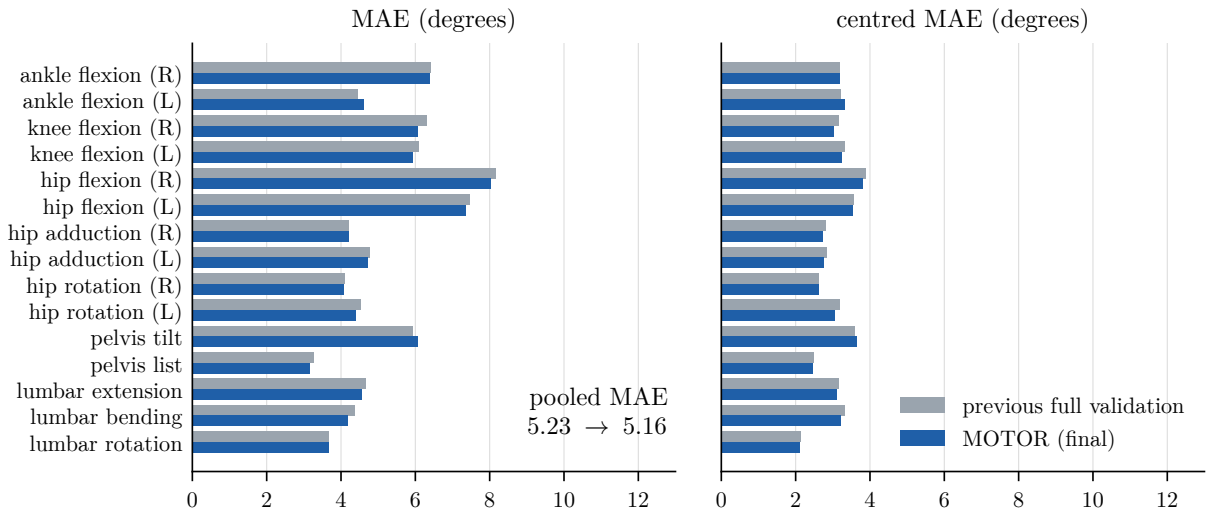


Figure 5: MOTOR's per-coordinate MAE and centred MAE on the matched set, beside MOTOR's previous full run of the same inputs (before changes to its contact, floor and knee ligaments), scored with the same code.

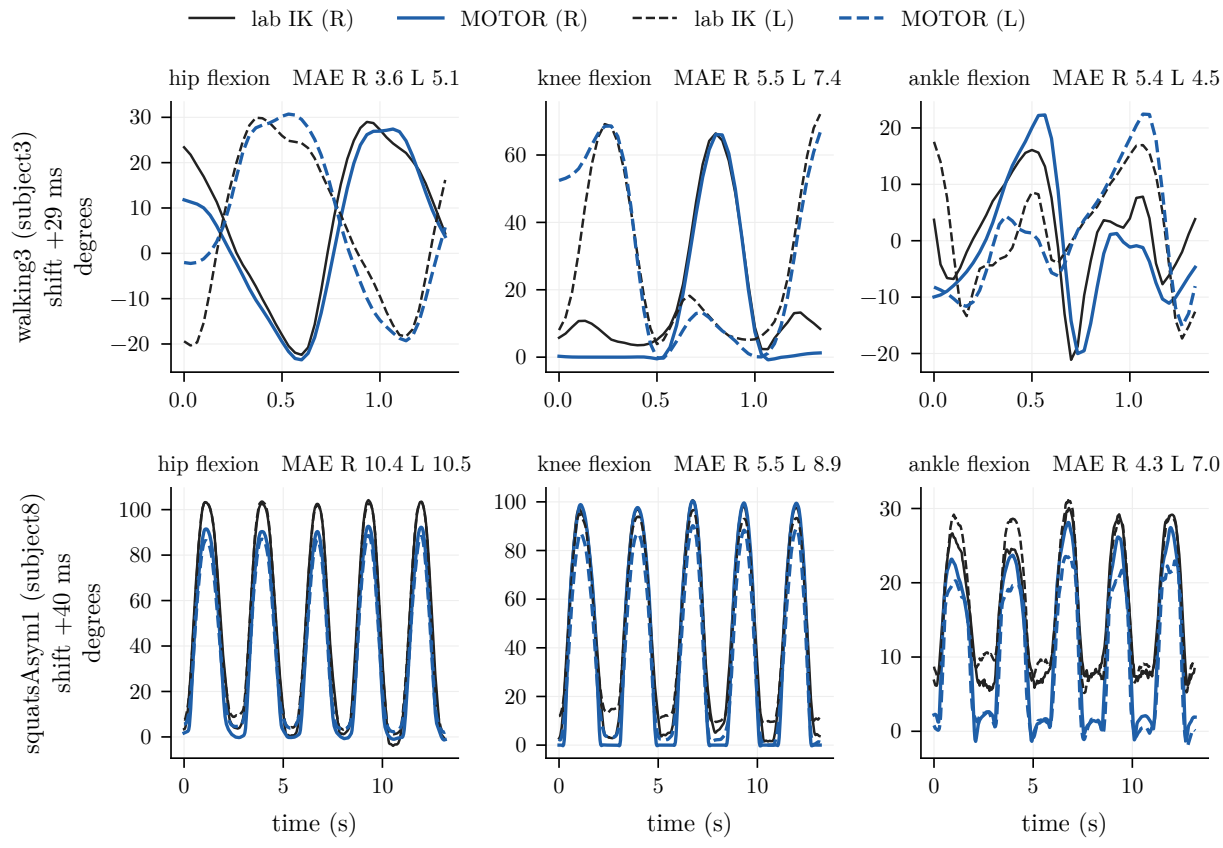


Figure 6: MOTOR's hip, knee and ankle flexion in the 45° camera's median walking and squat trial (by angle error) against the lab's inverse kinematics moved by the scored time shift.

C.2 Per subject group, activity and camera

Table 3: Per group: inputs, matched MAE and cMAE (degrees), vertical GRF MAE (% body weight, both feet), median vertical r per foot over the trials where it is defined, joint moment MAE over OpenCap's 15 (% weight × height), mean EMG r over the 16 channels. Cameras: LabValidation's numbering; Cam1 is the envelope's 45° camera.

Group	n	MAE	cMAE	GRF vert.	r R / L	Moments	EMG r
held out 7–11	396	5.13	3.02	16.28	0.77 / 0.79	2.41	0.24
training 2–5	319	5.62	3.54	16.15	0.78 / 0.77	2.42	0.22
walking	135	4.38	2.92	19.55	0.82 / 0.80	1.88	0.27
walking (trunk sway)	135	4.67	3.05	16.31	0.82 / 0.84	1.72	0.27
squats	87	5.79	3.26	6.32	0.53 / 0.47	1.58	0.17
sit-to-stand	90	5.39	2.92	11.58	0.68 / 0.80	1.55	0.20
drop jump	268	6.03	3.63	19.27	0.78 / 0.78	3.59	0.22
Cam0	144	5.21	3.14	14.95	0.75 / 0.83	2.44	0.23
Cam1 (45°)	141	5.62	3.33	15.50	0.73 / 0.81	2.54	0.22
Cam2	144	5.36	3.53	15.54	0.81 / 0.82	2.21	0.25
Cam3	143	5.35	3.20	15.96	0.81 / 0.70	2.44	0.23
Cam4	143	5.23	3.05	19.16	0.78 / 0.75	2.43	0.23

C.3 Ground reaction force

Table 4: Ground reaction force per activity: MAE in % body weight on three axes with both feet summed, as OpenCap scored it, and the share of the plate's stance frames on which MOTOR's foot carries force.

Activity	vertical	anterior	medio-lateral	3-axis mean	stance carried R / L
all	16.22	7.41	2.99	8.9	0.82 / 0.79
walking	19.55	12.03	3.02	11.5	0.82 / 0.75
walking (trunk sway)	16.31	10.86	3.49	10.2	0.83 / 0.85
squats	6.32	3.53	3.30	4.4	0.99 / 0.95
sit-to-stand	11.58	4.42	3.25	6.4	0.82 / 0.82
drop jump	19.27	5.61	2.54	9.1	0.76 / 0.71
OpenCap, 2 cameras	11.4	3.5	3.8	6.2	–

Per foot over all 715 inputs, the vertical force's MAE is 17.8% of body weight on the right and 17.2% on the left, and the centre of pressure, with each trial's constant offset removed, is 47.6 mm off in the anterior and 20.2 mm in the medio-lateral direction. Each stance's vertical impulse is 12.6% below the plate's on average.

C.4 Joint moments

In walking, the early-stance knee adduction moment's peak is overestimated: on the left knee MOTOR's mean is 3.5% of weight times height against the lab's 2.1%, a mean error of 1.5% against OpenCap's published 0.30%. In the recorded runs, the knee of the leg far from the camera sits 37 to 44 mm from the centre of pressure in the frontal plane where the lab has 9 to 16 mm, which accounts for most of this error.

C.5 Residuals

Table 5: Joint moment MAE (% weight × height) and r per coordinate over all 715 inputs, against OpenSim's inverse dynamics on the lab's own model. OpenCap's published mean over its 15: 1.2.

Coordinate	MAE	r	Coordinate	MAE	r
hip flexion r	4.06	0.53	hip flexion l	3.82	0.53
hip adduction r	1.97	0.21	hip adduction l	1.86	0.16
hip rotation r	0.80	0.11	hip rotation l	0.76	0.13
knee angle r	2.83	0.39	knee angle l	2.71	0.37
ankle angle r	2.00	0.48	ankle angle l	2.01	0.36
subtalar angle r	0.65	0.43	subtalar angle l	0.57	0.40
lumbar extension	5.39	0.38	elbow flex r	0.19	0.40
lumbar bending	3.62	0.27	elbow flex l	0.23	0.43
lumbar rotation	3.14	0.27	pro sup r / l	0.02 / 0.02	0.10 / 0.10
OpenCap's 15, mean	2.41				

Table 6: The pelvis residual per group: the median over trials of each trial's median force (% body weight) and moment (% weight × centre-of-mass height), the median of each trial's 95th-percentile force, and the mean share of frames within both limits (2% body weight, 1% weight × centre of mass height). No trial is within them on every frame.

Group	trials	force	moment	force, 95th pct.	frames within
all	715	7.0	8.90	24.4	21%
held out 7–11	396	7.1	9.31	23.9	21%
walking	135	7.8	10.99	23.0	6%
walking (trunk sway)	135	7.0	9.59	20.3	7%
squats	87	0.7	1.06	8.4	49%
sit-to-stand	90	0.2	2.82	8.8	46%
drop jump	268	10.9	9.32	85.5	17%

C.6 Muscle activations against EMG

Table 7: Each EMG channel against MOTOR's activation of that muscle, all 715 inputs: the share of frames on which both are on or both off, Pearson's r, and the lag (s) at the cross-correlation peak.

Muscle	on/off	r	lag	Muscle	on/off	r	lag
bflh l	0.65	0.13	-0.130	semiten r	0.70	0.23	-0.034
bflh r	0.65	0.13	-0.144	soleus l	0.70	0.26	-0.030
gasmed l	0.70	0.23	-0.009	soleus r	0.69	0.28	-0.041
gasmed r	0.69	0.26	-0.022	tibant l	0.59	0.22	+0.031
glmed1 l	0.51	0.13	+0.051	vaslat l	0.72	0.40	-0.003
glmed1 r	0.52	0.15	+0.040	vaslat r	0.72	0.42	-0.018
recfem l	0.56	-0.08	-0.002	vasmed l	0.72	0.36	-0.024
semiten l	0.66	0.18	-0.053	vasmed r	0.76	0.41	-0.033

Table 8: OpenCap's vasti symmetry index in squats, per camera: the AUC of MOTOR's index against the asymmetry the instructions asked for and against the EMG's own label (index over 1.15). OpenCap: 0.83 against the EMG label.

Camera	AUC, instructed	AUC, EMG label
Cam0	0.84	0.75
Cam1	0.72	0.62
Cam2	0.95	0.84
Cam3	0.90	0.83
Cam4	0.81	0.48
all	0.73	0.66

C.7 Time per step

Table 9: MOTOR's wall seconds per step on one NVIDIA L4, over the 715 runs: median, range, and the median per frame (clips of 30 Hz).

Step	median s	min s	max s	s per frame
fit	21.825	14.483	623.317	0.303
set-up: physics model, GPU arrays, filtered path	4.697	3.878	32.851	0.061
footing	4.669	1.758	35.312	0.062
residual reduction, iterations	19.862	5.910	392.306	0.246
residual reduction, tracking and per-frame caps	4.603	1.714	45.888	0.061
checks: residual record, feet on the floor, tracking	2.120	1.111	17.879	0.028
static optimisation, stage 1 (ground reaction)	3.003	0.114	28.318	0.034
static optimisation, stage 2 (joint limits and muscles)	2.296	1.169	22.356	0.032
inverse dynamics, contacts and muscle paths	1.120	0.356	12.459	0.013
writing records and force measures	0.264	0.042	2.639	0.003
saving arrays	0.005	0.003	0.012	0.000
total	66.214	34.843	788.668	0.871

Table 10: Wall time per trial, MOTOR against published solves. OpenCap's solves are from its LabValidation release (80 muscles, windows of 0.99 to 1.68 s, solve only, hardware not recorded).

Method	Clip	Wall time
OpenCap solve [15]	walk, drop jump, squat	4196, 4096, 2635 s
Moco, MocoTrack [12]	gait	about 130 min
OpenSim CMC [38]	gait	about 10 min
MOTOR, one L4	LabValidation, 715 trials	66 s median (35–789 s)

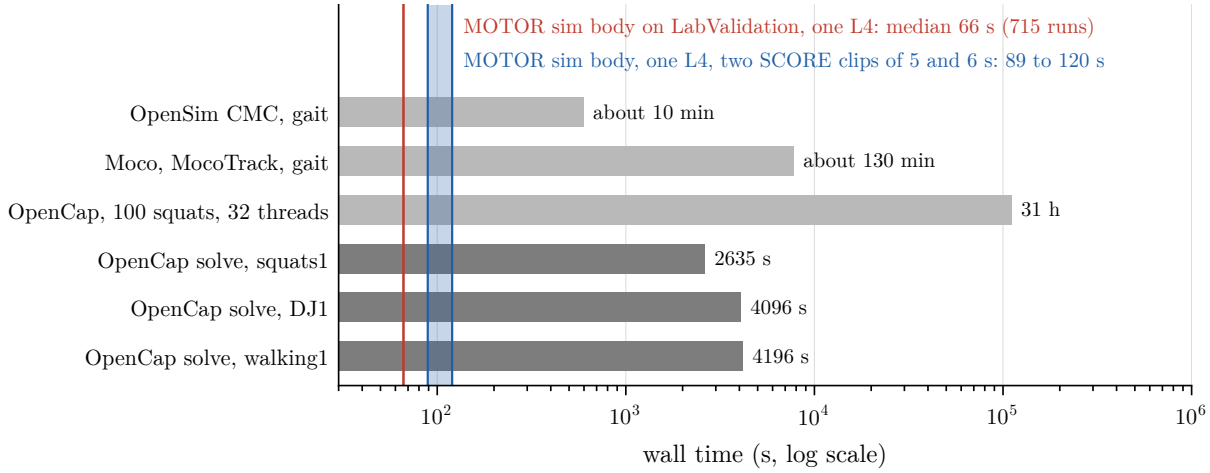


Figure 7: MOTOR's wall time per trial on a log scale: OpenCap's recorded solves and published OpenSim times (grey), MOTOR on two SCORE clips (blue band) and its median on LabValidation (red).

C.8 What the run shows

What the input carries. MOTOR follows its input, so the input's errors pass through. The elbows come out straighter than the lab's in every one of the 89 matched trials, by a mean 21.1° (right) and 23.7° (left), with centred errors of 3.0 and 4.6°. On 540 walking arm trials GEM-X's own elbow is a median 18.5° straighter than the markers', and the fit's elbow differs from GEM-X's by a median 2.5°. The pelvis tilt is offset in one direction in 82 of 89 matched trials, by a mean 5.3°. Neither is calibrated away: no angle is fitted to the reference, and reading each subject's own standing trial as a static calibration, tried on the held-out subjects, made their pelvis tilt, elbows and matched error worse.

Forces, residuals and the two sides. The pelvis residual stays above its limits, the anterior ground reaction force error is about twice OpenCap's, and the activation pattern is partly off (rectus femoris does not follow its EMG, $r = -0.08$). The ground force is split between the feet by the skin points that reach the floor and by the pelvis' balance, so a foot that GEM-X places a little higher carries less: in the squat of Figure 1, at its deepest frame, the left foot carries 90% of body weight and the right 53%, and the calf muscles follow that split rather than the symmetric movement.

Camera sampling and fast movements. GEM-X resamples every clip to 30 Hz, so an impact or a push-off is sampled at that rate whatever the phone recorded. Of the five activities the drop jumps have the highest joint moment error (3.59% of weight times height, against 1.55 to 1.88 for the others), the largest pelvis residual (a median of 10.9% of body weight, its 95th percentile 85.5%), and two of the five inputs that failed did so because a drop jump frame's fit did not converge.

The fit's own records. The calibration (level 1) tunes on synthetic SOMA bodies, not on people. The last calibration run whose held-out score is recorded met its rule: its check bodies' score fell from 4.26 to 3.56 while the tuning bodies' fell from 4.53 to 3.44, and the rule's comparison, 3.56 against the tuning bodies' 3.51, was within 10%. The tuned point the fit uses now came from a later run with a broader objective, whose held-out score is not yet recorded. The bones are not yet all inside the skin: on the fifteen clips of the synthetic body of mean shape, 5.15% of the bone points lie outside it (averaged over body regions), against a target of under 5%.

C.9 Failures

Table 11: The 5 of 720 inputs without a score, by cause.

Cause	Inputs
The GPU fit did not stop in 3000 iterations (subject 10, squats1, two cameras)	2
A frame's fit did not converge in 300 steps (two drop jumps)	2
Passed its job's time limit (subject 10, squats1, a third camera)	1
Total	5

D Model and solver parameters

Table 12: The bone groups: each SOMA segment, the MyoFullBody bone that carries its rotation, and the bones the group covers. Every sided group exists on both sides; 7 middle and 7 sided groups make 21, of which 20 drive their frame bone.

SOMA segment	Frame bone	Bones in the group
Hips	pelvis	pelvis, sacrum
Spine1	lumbar 4	lumbar 5 and 4
Spine2	lumbar 1	lumbar 3 to 1
Chest	torso	torso, thorax, cervical spine
Neck1	neck	neck
Neck2	neck (drives nothing)	–
Head	head	head
Shoulder	scapula	clavicle, scapula
Arm	humerus	humerus
ForeArm	ulna	ulna, radius
Hand	lunate	carpals, metacarpals, phalanges
Leg	femur	femur
Shin	tibia	tibia
Foot	calcaneus	talus, calcaneus
ToeBase	toes	toes

Table 13: MOTOR's parameters in one place.

Part	Parameter	Value
Model	muscles, bones, joints	416, 86, 132
	joints: root, coupled, free	1, 36, 95
Landmarks	defined, used by the fit	102, 84
Fit	markers' goal λ_m	2.1169
	skin goal λ_s	10^3
	most steps per frame	300
Level 1	segment end's reach	40 mm
	held-out rule	check bodies within 10% of tuning bodies
Level 2	sampled frames	about 30
	landmark's and inside joint's reach	3 cm
Forces	contact gap	5 mm
	a seen standing foot pushed up to	5 cm over the floor
	ground share weight, rounds	0.03 per newton, 2
	smoothing cut-off	6 Hz
	residual reduction: rounds	3
	per-frame cap	5° per segment, 2 cm at the pelvis
	mass change	at most 2%
	residual limits	2% body weight; 1% weight × centre of mass height

E Disclosure, data and code

The code of MOTOR and of its validation was written by a coding agent (Claude Code) under the author's direction and review; this draft was written the same way. No result was hand-picked. The only settings chosen on LabValidation (MOTOR's contact gap, the height from which a seen standing foot is pushed, the weight that shares the ground force over the touching points, and the clearance of a drop jump's box) were chosen on subjects 2 to 5 against their force plates, never

against joint angles; subjects 7 to 11 were never used to choose any setting. Of these, only the standing foot's height changed from its first value, from 30 to 50 mm.

MOTOR, its validation tools and this paper's sources are open under the MIT licence at https://github.com/Babon-Innovations-b-v/score; every number in this paper is read from a file or record named in a comment beside it in the paper's source. We collected no new data from people. LabValidation is the public release of OpenCap's authors [15], used under its Apache-2.0 licence and fetched on demand, never redistributed.